

Final Path Experiment — Results Analysis

Date: 2026-06-08
Setup: fly_threshold_10km (10 km fly/stationary threshold)
Split: year-grouped 80/20 (train on earliest 80% of years, test on latest 20%)
k values: 7, 14, 30
Epochs: max 200, patience 30

Model Architecture Summary

Model	Family	Layers	Hidden Dim	Params (k=7)
Direct MLP	direct	0	—	25K
Direct Transformer 4L	direct	4	d_model=128	805K
Triline LSTM 4L	triline	4	hidden=256	2.2M
Triline Transformer V2	triline	4	d_model=256, 8 heads, CLS token	3.3M

All models use **no bird embeddings** (removed for year-grouped split compatibility).
Triline models predict: fly/stationary (BCE), log distance (MSE), and direction sin/cos (MSE).
Direct models predict: lat/lon delta only (MSE).

With Weather

k=7

Model	Mean km	Median km	P90 km	Fly Recall	Best Epoch
Direct MLP	19.02	4.85	59.98	—	9
Direct Transformer 4L	17.79	3.69	56.87	—	12
Triline LSTM 4L	17.86	2.41	53.54	0.615	4
Triline Transformer V2	17.96	2.41	57.60	0.658	17

Winner: LSTM 4L (better P90, faster convergence). TransV2 matches on median but trails on P90.

k=14

Model	Mean km	Median km	P90 km	Fly Recall	Best Epoch
Direct MLP	20.68	5.13	66.20	—	5
Direct Transformer 4L	21.57	8.22	60.16	—	6
Triline LSTM 4L	18.69	2.74	59.06	0.679	4

Model	Mean km	Median km	P90 km	Fly Recall	Best Epoch
Triline Transformer V2	18.85	2.51	61.36	0.684	30

Winner: LSTM 4L (better mean, median, P90). TransV2 wins fly recall by 0.005.

k=30

Model	Mean km	Median km	P90 km	Fly Recall	Best Epoch
Direct MLP	19.08	5.02	58.84	—	6
Direct Transformer 4L	17.47	3.45	59.76	—	7
Triline LSTM 4L	17.03	2.47	51.57	0.764	2
Triline Transformer V2	17.33	2.42	52.78	0.670	10

Winner: LSTM 4L (dominant fly recall, better P90).

Without Weather

k=7

Model	Mean km	Median km	P90 km	Fly Recall	Best Epoch
Direct MLP	19.21	4.47	62.18	—	11
Direct Transformer 4L	17.65	2.47	58.41	—	13
Triline LSTM 4L	18.19	2.26	63.06	0.724	9
Triline Transformer V2	18.18	2.56	62.39	0.696	23

Winner: LSTM 4L (better median, fly recall).

k=14

Model	Mean km	Median km	P90 km	Fly Recall	Best Epoch
Direct MLP	20.63	5.14	64.67	—	8
Direct Transformer 4L	19.18	3.69	62.64	—	11
Triline LSTM 4L	18.85	2.49	62.29	0.731	8
Triline Transformer V2	18.66	2.54	56.66	0.738	28

Winner: TransV2 (best mean, P90, fly recall). LSTM 4L wins median by 0.05 km.

k=30

Model	Mean km	Median km	P90 km	Fly Recall	Best Epoch
Direct MLP	20.84	5.75	62.48	—	4
Direct Transformer 4L	18.22	4.35	57.93	—	11
Triline LSTM 4L	17.06	2.45	49.66	0.702	7
Triline Transformer V2	17.20	2.27	49.34	0.718	15

Winner: Tie. TransV2 wins median + P90 + fly recall. LSTM 4L wins mean.

Cross-K Trends

Trend	Observation
k ↑ → better median	All models improve median error as lookback increases. LSTM 4L: 2.41→2.74→2.47 (weather). TransV2: 2.41→2.51→2.42.
k ↑ → TransV2 gains on LSTM	At k=7 LSTM wins. At k=14 close. At k=30 TransV2 competitive or winning.
Convergence speed	LSTM 4L converges 2-4× faster than TransV2 (best epoch 2-9 vs 10-30).
P90 variance	TransV2 P90 is more stable across k (52-62 range) vs LSTM (49-63 range).

Weather Impact

Weather effect measured as `mean_error_km_weather` - `mean_error_km_noweather` (negative = weather helps):

Model	k=7	k=14	k=30
Triline LSTM 4L	-0.33	-0.16	-0.02
Triline Transformer V2	-0.21	+0.19	+0.14

Weather impact is small and inconsistent. At k=7 weather helps both models slightly. At k=14,30 the effect oscillates. The year-grouped split may explain this: weather patterns differ between train and test years, limiting the predictive value. Weather features were more useful under the original chronological split where train and test shared the same year's climate.

Parameter Efficiency

Model	Params	Mean Error (k=30, weather)	Error per M param
Direct MLP	78K	19.08	244.6
Direct Transformer 4L	808K	17.47	21.6
Triline LSTM 4L	2.2M	17.03	7.7

Model	Params	Mean Error (k=30, weather)	Error per M param
Triline Transformer V2	3.3M	17.33	5.3

Direct MLP is the most parameter-efficient per unit of error. Triline models gain modestly for large parameter increases.

Conclusions

1. **Triline LSTM 4L is the overall winner.** It wins or ties 9 of 12 metric slices across k values and weather modes. It converges 2-4× faster than the transformer and has fewer parameters (2.2M vs 3.3M).
2. **Triline Transformer V2 is competitive at k=30.** It wins P90 (49.34 vs 49.66 without weather) and median (2.27 vs 2.45) at the longest lookback. The transformer architecture benefits from longer sequences — at k=30 attention has enough temporal range to be useful.
3. **Direct models are surprisingly strong.** Direct Transformer 4L (805K params) matches or beats Triline models on median error at k=7,14 without weather. This architecture is simpler and faster to train.
4. **Weather effect is weak under year-grouped split.** Under the chronological split, weather provided a consistent benefit. Under year-grouped split, different years have different weather patterns, so weather features don't generalize as well across train/test years.
5. **LSTM wins because inductive bias matches the task.** Bird trajectory prediction at $k \leq 30$ is inherently sequential: the most recent day is the strongest predictor. LSTM's last-hidden-state recency bias is the right inductive bias. The transformer's global attention is more flexible but needs more data and longer sequences to earn its cost.

Rollout Evaluation

Autoregressive rollouts use a minimum 60-day observed context and extend that context until observed displacement reaches 50 km when possible. Each model then rolls forward for the rest of the path using its trained trailing context window.

Model	With weather mean km	Without weather mean km	With - without mean km	With weather final km	Without weather final km
Direct	417.22	240.07	177.15	603.58	343.52
LSTM	165.16	175.34	-10.18	193.77	206.95
Transformer	237.28	180.84	56.43	323.68	227.91

Valid rollout paths: with weather 27, without weather 27. Skipped paths: with weather 7, without weather 7.

Weather slightly worsens average rollout error across the selected model families.

Artifacts:

- Per-step and per-path errors: D:\UCSD\ECE 228\ECE-228-Project\rollout\with_weather, D:\UCSD\ECE 228\ECE-228-Project\rollout\without_weather
- Weather comparison CSV: D:\UCSD\ECE 228\ECE-228-Project\rollout\rollout_error_comparison.csv
- Representative GIFs: D:\UCSD\ECE 228\ECE-228-Project\rollout\with_weather\gifs, D:\UCSD\ECE 228\ECE-228-Project\rollout\without_weather\gifs